

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5, BL6)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

CONSTRAINT-BASED PROBLEM SOLVING

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
CONSTRAINT-BASED PROBLEM SOLVING

1. A financial institution must secure all web transactions using TLS, but legacy systems only support older SSL versions. Design a solution to ensure secure communication.

Constraints:

- A. Backward compatibility must be maintained for 20% legacy clients
- B. Sensitive data must not be transmitted using SSL
- C. System downtime must be minimal during migration
- D. Cost of implementation should be optimized

2. An e-commerce platform is considering implementing the SET protocol for payment security but currently uses TLS-based encryption. Analyze and propose a decision.

Constraints:

- A. Must support high transaction throughput (10,000 transactions/minute)
- B. Compatibility with modern browsers is required
- C. Implementation complexity should be minimized
- D. Regulatory compliance (PCI-DSS) must be ensured

3. A corporate network is experiencing repeated malware infections that spread rapidly without user interaction. Propose a containment and prevention strategy.

Constraints:

- A. The network consists of 500+ interconnected systems
- B. Downtime per system should not exceed 10 minutes
- C. Detection must occur within 2 minutes of infection
- D. Solution must differentiate between virus, worm, and Trojan behavior

4. An organization needs to deploy a firewall architecture to protect its internal network from external threats and insider attacks. Design the solution.

Constraints:

- A. Must support both packet filtering and application-level inspection
- B. Latency introduced should not exceed 5 ms per request
- C. Should handle at least 1 Gbps traffic
- D. Must allow secure remote access for employees

5. A large enterprise network generates excessive false positives in its IDS, leading to alert fatigue. Propose an optimized IDS/IPS solution.

Constraints:

- A. Detection accuracy must be at least 95%
- B. False positive rate should be reduced below 5%
- C. System must process real-time traffic of 2 Gbps
- D. Solution should integrate with existing firewall infrastructure

RUBRICS FOR CALCULATION:

Q. No	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

CO5 AT1 CRYPTO:

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1. TLS Migration for Legacy SSL Clients

Solution: Implement a **TLS-first migration using a secure gateway/load balancer**.

- Enable **TLS 1.2/1.3** for all modern clients.
- Keep SSL support only for the **20% legacy clients** through an isolated legacy gateway.
- Do **not allow sensitive data** to pass through SSL; legacy users can access only limited/non-sensitive services.
- Gradually upgrade or replace legacy clients.
- Use monitoring to identify remaining SSL users and migrate them.
- Perform migration in phases to keep downtime minimal.
- Reuse existing servers/load balancers where possible to reduce cost.

Architecture:

Users → Load Balancer → TLS 1.2/1.3 → Application

Legacy Users → Secure Gateway → Restricted Services

Result: Secure TLS communication is provided while maintaining limited backward compatibility without exposing sensitive information over SSL.

2. SET vs TLS for E-Commerce

Recommendation: Continue with TLS rather than implementing SET.

Factor	TLS	SET
Modern browser support	✓ Excellent	✗ Poor/limited
Throughput	✓ High	✗ More complex
Implementation	✓ Simple	✗ Complex
PCI-DSS	✓ Can support compliance	✓ Designed for payment security

Factor	TLS	SET
Current infrastructure	✓ Already used	✗ Requires major changes

Solution:

- Continue using **TLS 1.2/1.3** for secure communication.
- Use a PCI-DSS-compliant payment gateway.
- Tokenize payment information where possible.
- Encrypt sensitive information in transit and at rest.
- Apply strong authentication, logging, access control, and regular security testing.

Result: TLS provides better browser compatibility, lower implementation complexity, and sufficient performance for **10,000 transactions/minute** when properly engineered.

3. Malware Containment and Prevention

Solution: Deploy a combination of **EDR + IDS/IPS + network segmentation + automated isolation**.

Detection:

- **Virus:** Detects malicious file infection and replication.
- **Worm:** Detects rapid network-to-network propagation without user interaction.
- **Trojan:** Detects suspicious applications/processes disguised as legitimate software.

Process:

System → EDR Monitoring → IDS/IPS → Threat Detection → Automatic Isolation → SOC Alert

Actions:

1. Monitor all 500+ systems continuously.
2. Detect suspicious behavior within **2 minutes**.
3. Automatically isolate the infected machine from the network.
4. Block malicious IPs, domains, files, and processes.
5. Scan and clean the affected system.

6. Restore connectivity after verification.
7. Use VLAN/network segmentation to prevent widespread propagation.

Result: Automated isolation limits malware spread while keeping individual-system downtime below **10 minutes**.

4. Firewall Architecture

Solution: Use a **multi-layer firewall architecture** with packet filtering, stateful inspection, application inspection, and secure remote access.

Architecture:

Internet → Perimeter Firewall → DMZ → Internal Firewall → Internal Network

Remote Employees → VPN Gateway → Firewall → Internal Network

Components:

- **Packet filtering:** Blocks unauthorized IP addresses, ports, and protocols.
- **Stateful firewall:** Tracks active connections.
- **Application-level inspection:** Examines HTTP, HTTPS and other application traffic.
- **DMZ:** Hosts public-facing web/mail servers separately.
- **VPN:** Provides encrypted remote access for employees.
- **IDS/IPS:** Detects and blocks suspicious traffic.

Performance:

- Use a high-performance firewall capable of **≥1 Gbps** throughput.
- Use hardware acceleration and optimized rules to maintain latency below **5 ms**.
- Apply firewall rules from most specific to general to reduce processing time.

Result: The architecture protects against both external and internal threats while supporting secure remote access.

5. Optimized IDS/IPS

Solution: Deploy a **hybrid IDS/IPS with behavioral detection, signature detection, and firewall integration**.

Architecture:

Network Traffic → IDS/IPS → Detection Engine → Risk Analysis → Firewall

Optimization techniques:

- Use updated attack signatures.
- Apply **behavior-based detection** for unknown attacks.
- Use machine-learning/anomaly detection to identify unusual traffic.
- Assign risk scores to alerts.
- Correlate multiple alerts before generating a high-priority alert.
- Automatically send confirmed malicious IPs to the firewall for blocking.
- Tune rules regularly to eliminate unnecessary alerts.

Performance requirements:

- Use load balancing or multiple sensors to handle **2 Gbps** traffic.
- Optimize signatures and packet inspection.
- Target **≥95% detection accuracy**.
- Keep the false-positive rate **below 5%**.

Result:

Traffic → IDS/IPS → Filter/Analyze → High Confidence Threat → Firewall Block

This reduces alert fatigue while maintaining real-time protection and integrating the IDS/IPS with the existing firewall.